

? FUNCTION DECLARATION (BASICS)

1. Write a function that prints **"Hello JavaScript"**.
 2. Write a function that takes two numbers and prints their **sum**.
 3. Write a function that checks whether a number is **even or odd**.
 4. Write a function that finds the **square of a number**.
 5. Write a function that checks whether a person is **eligible to vote**.
 6. What happens if you call a function **before its declaration**?
 7. Write a function to return the **largest of two numbers**.
 8. Write a function to convert **Celsius to Fahrenheit**.
 9. Write a function that prints numbers from **1 to 10**.
 10. Write a function that calculates **simple interest**.
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? PARAMETERS & ARGUMENTS (LOGIC BASED)

11. Write a function that takes **name and age** and prints a message.
 12. Write a function that accepts **marks of 3 subjects** and returns total.
 13. What is the difference between **parameters and arguments**? Explain with example.
 14. Write a function that takes a number and checks **positive, negative, or zero**.
 15. Write a function that takes **length and breadth** and returns rectangle area.
 16. What happens if fewer arguments are passed than parameters?
 17. What happens if more arguments are passed than parameters?
 18. Write a function that takes salary and bonus percentage and returns final salary.
 19. Write a function that takes radius and returns **circle area**.
 20. Write a function that checks whether a year is a **leap year**.
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❓ RETURN STATEMENT (VERY IMPORTANT)

21. Write a function that **returns** sum of two numbers.
22. What happens if a function has **no return statement**?
23. Predict the output:

```
function test() {  
  return;  
}  
console.log(test());
```

24. Can a function return **multiple values**? How?
 25. Write a function that returns **Pass or Fail** based on marks.
 26. What happens to code written **after return**?
 27. Write a function that returns **true if number is prime**, else false.
 28. Write a function that returns the **factorial** of a number.
 29. Write a function that returns **maximum of three numbers**.
 30. Write a function that returns **grade** based on percentage.
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🌀 FUNCTION EXPRESSION

31. Convert a function declaration into a **function expression**.
32. Can you call a function expression **before definition**? Why?
33. Predict the output:

```
console.log(add(2,3));  
var add = function(a,b){ return a+b };
```

34. Write a function expression to check **even or odd**.
 35. Write a function expression that returns **square of a number**.
 36. What is the difference between function declaration and function expression?
 37. Write a function expression that calculates **discounted price**.
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🔍 ARROW FUNCTIONS (BASIC)

38. Convert this into arrow function:

```
function add(a,b){ return a+b; }
```

39. Write an arrow function that returns **cube of a number**.

40. Write an arrow function with **single parameter** (no brackets).

41. Write an arrow function with **implicit return**.

42. Predict the output:

```
const fn = () => { value: 10 };  
console.log(fn());
```

43. Why does the above return undefined?

44. Write an arrow function to check **eligibility for exam**.

45. Can arrow functions be used as constructors? Why?

🎯 DEFAULT PARAMETERS & SCENARIOS

46. Write a function with default parameter for **tax = 18%**.

47. What happens if default parameter is not used?

48. Predict the output:

```
function test(a = 5, b = a * 2) {  
  return a + b;  
}  
console.log(test());
```

49. Write a function that calculates **final bill** using default discount.

50. Explain a real-world scenario where **default parameters** are useful.

✅ INSTRUCTIONS

- Solve all questions by writing code & logic
- Do not run code initially
- After completion, say:

👉 "Give me answers for Functions questions"

Java Script Functions (logic, Scenarios, Tricky)

I will then provide: - Step-by-step solutions - Common mistakes - Interview tips - Optimized versions

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